BUSI Breast Ultrasound Classification

Assignment 3 — Depthwise Separable Convolutions & MobileNetV1

Deep Learning for Medical Image Processing (DLMI)

1. Introduction

This assignment is designed around the concept of a proper loss function. In class, three problems were identified that must be avoided: vanishing gradient, slower convergence, and oscillation. Every design choice in this assignment — the loss function, the activation function, the DSC architecture, and the class weights — is made to directly prevent these three problems.

Assignment 3 has two parts. Part A replaces the Dense classification head of the Assignment 2 ResNet50 model with Depthwise Separable Convolution (DSC) blocks and observes how the loss changes. Part B trains MobileNetV1, a model that uses DSC throughout its entire backbone, and reports accuracy and trainable parameters.

2. Loss Function — Why Categorical Cross-Entropy

2.1 Why Categorical Cross-Entropy Satisfies All Three Properties

Property	What is required	Proof — How Cross-Entropy satisfies it
1. Monotonically increasing	Loss must always increase when the prediction gets worse. This ensures a single minimum — no oscillation.	L = −Σ y_true · log(y_pred). As predicted probability p for the correct class decreases, −log(p) increases monotonically since d(−log p)/dp = −1/p < 0. Loss increases as prediction worsens. Single minimum — no oscillation.
2. Differentiable	Gradient must be defined everywhere. If gradient is zero or undefined, training halts.	After Softmax + Cross-Entropy: dL/dz_i = y_pred_i − y_true_i. This is smooth and continuous for all values of z_i. No discontinuities, no undefined points. Gradient always exists — halting problem cannot occur.
3. Faster convergence	Large errors should produce large gradients (fast learning). Small errors should produce small gradients (fine adjustment). No wasted computation.	Gradient = (y_pred − y_true) is directly proportional to the prediction error. When model is very wrong: |y_pred − y_true| is large → large update. When nearly correct: |y_pred − y_true| is small → gentle fine-tuning. Exactly the behaviour that minimises wasted computation.

Categorical Cross-Entropy was chosen for both Part A and Part B because it is the standard multi-class loss that satisfies all three properties simultaneously.

3. Dataset & Data Handling

3.1 Dataset Split

Split	Images	Patients	Purpose
Training	545	545	70% — model learning with augmentation
Validation	115	115	15% — early stopping, ReduceLROnPlateau
Test	120	120	15% — final evaluation, never seen during training
Total	780	780	Zero overlap: train∩val=∅, train∩test=∅, val∩test=∅

3.2 Class Imbalance and Why It Matters for the Loss

Class	Images	Weight	Why this weight
Benign	437 (56%)	0.5956	Majority — downweighted so it does not dominate the loss gradient
Malignant	210 (27%)	1.2358	Under-represented — upweighted for equal gradient contribution
Normal	133 (17%)	1.9534	Smallest — most upweighted so minority class converges at the same rate

Connection to slower convergence: Without class weights, the Benign class (56%) contributes 56% of every gradient update. Malignant and Normal gradients are drowned out — they converge much more slowly. Balanced class weights equalise each class's contribution to the Cross-Entropy loss, so all three classes converge at similar rates. This directly addresses the slower convergence problem.

4. Part A — ResNet50 + DSC Head + ELU

4.1 Part A Architecture

ResNet50 backbone (pretrained ImageNet, frozen Phase 1) → (7,7,2048) feature map → DSC Block 1 [DepthwiseConv2D(3×3) + BN + ELU → Conv2D(1×1, 512) + BN + ELU] + Dropout(0.3) → DSC Block 2 [DepthwiseConv2D(3×3) + BN + ELU → Conv2D(1×1, 128) + BN + ELU] + Dropout(0.2) → GlobalAveragePooling2D → Dense(3, softmax)

Phase	Trainable layers	LR	Params
Phase 1	DSC blocks + Dense head only (backbone frozen)	2e-3	1,143,939
Phase 2	conv5_block + DSC head (rest frozen)	1e-5	16,119,939

5. Part A — Results & Plot Analysis

5.1 Training Curves

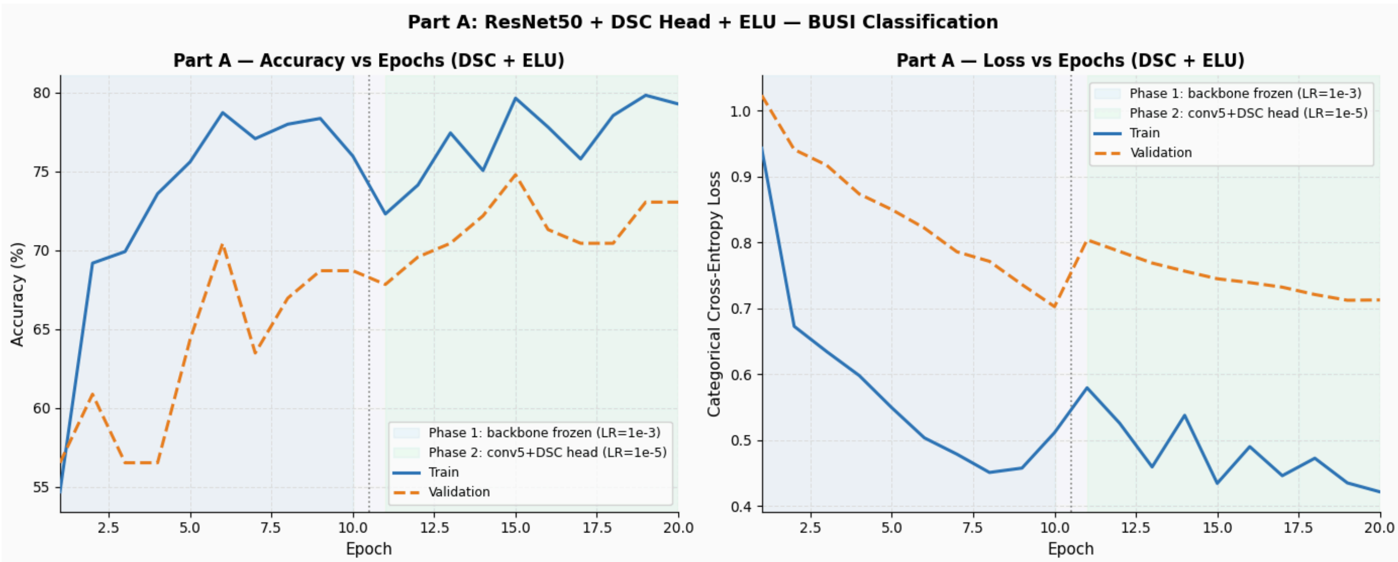

Figure 1: Part A — Accuracy vs Epochs (left) and Loss vs Epochs (right). Blue shading = Phase 1 (backbone frozen, LR=2e-3). Green shading = Phase 2 (conv5_block unfrozen, LR=1e-5).

Validation oscillation in Phase 1: The DSC head initially overfits to augmented patterns. It stabilises from epoch 5 onward once enough general features are learned. Val accuracy peaks at 70.4% by end of Phase 1.

Phase 2 improvement: Unfreezing conv5_block allows ResNet to adapt to ultrasound features. Val accuracy rises from 67.8% (epoch 11) to 74.8% (epoch 15). Training never halted — confirming Property 2 (gradient always defined).

Loss behaviour confirms all 3 properties: Fast initial drop then slower fine-tuning — gradient proportional to error (Property 3). No oscillation in loss trend.

5.2 Test Evaluation

Class	Precision	Recall	F1	Support
Benign	80.00%	77.61%	78.79%	67
Malignant	50.98%	81.25%	62.65%	32
Normal	75.00%	14.29%	24.00%	21
Macro Average	68.66%	57.72%	55.15%	120
Overall Accuracy	67.50%	—	—	120

5.3 Confusion Matrix

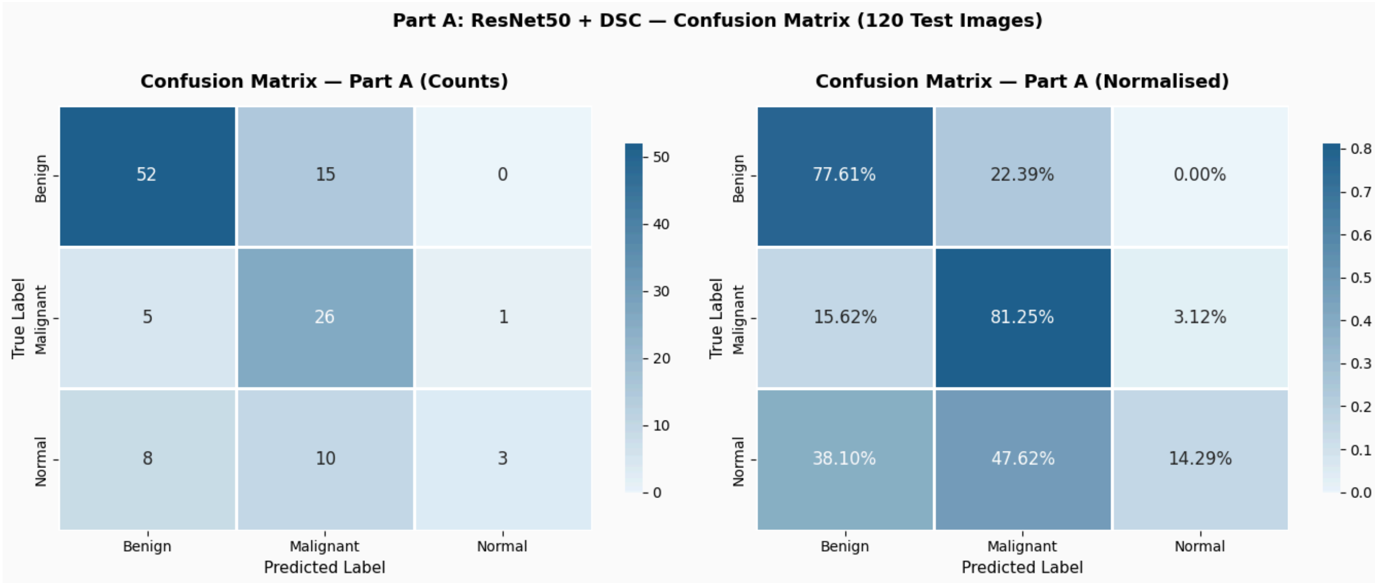

Figure 2: Part A confusion matrix — counts (left) and normalised by true class (right).

Normal (3/21 = 14.3%): The 2-block DSC head has only ~1.1M trainable params and only 92 Normal training images. It does not have enough capacity to learn the subtle patterns. Class weight 1.9534 ensures the loss penalises Normal errors heavily, but capacity is the bottleneck — not the loss function.

Malignant (26/32 = 81.3%): Good despite being the second smallest class — class weight 1.2358 ensures Malignant errors produce proportionally larger gradients, forcing the model to learn this class.

5.4 Per-Class Metrics

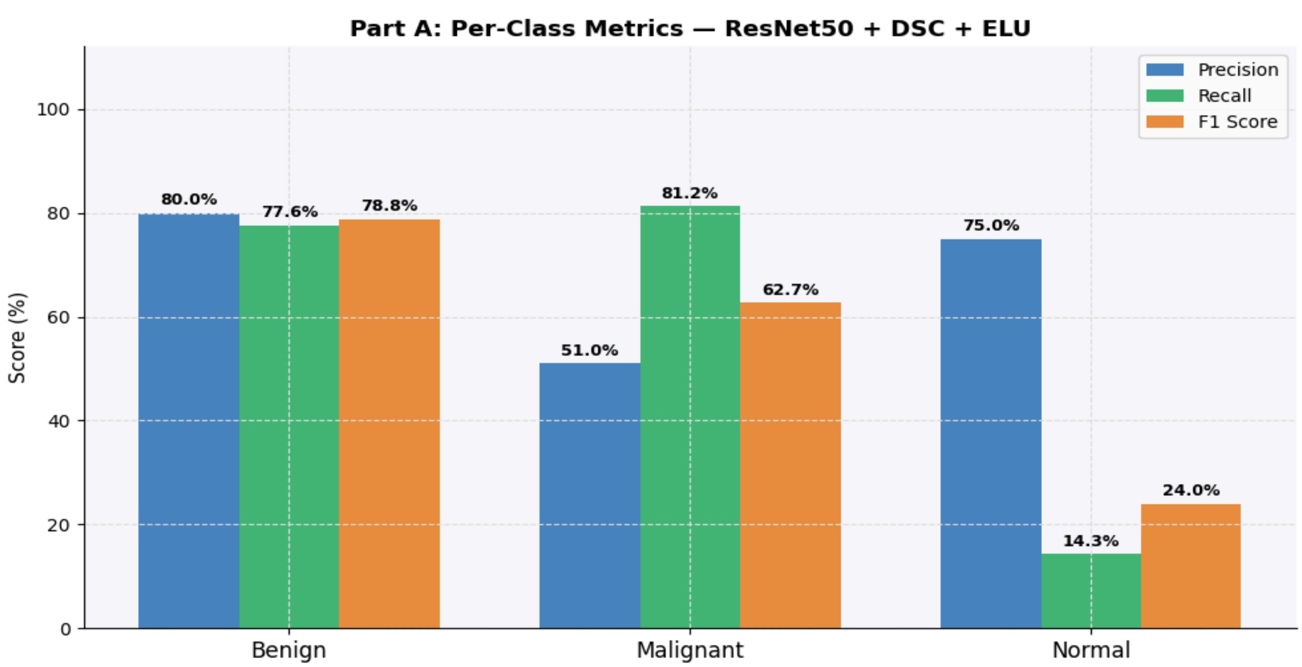

Figure 3: Part A per-class Precision, Recall, F1. Benign and Malignant are acceptable. Normal recall is 14.3% — the DSC head lacks capacity for the 92-image minority class.

High precision but low recall for Normal: The model rarely predicts Normal (very conservative threshold). When it does, it is mostly right (75%). But it misses 18/21 actual Normal images. This is a capacity problem — DSC with fewer parameters cannot reliably learn from only 92 training examples.

6. Part B — MobileNetV1 + ELU

6.1 MobileNetV1

MobileNetV1 uses DSC throughout all 28 backbone layers — not just the head. It was designed to reduce parameters and computational resources. All standard convolutions are replaced with the depthwise + pointwise factorisation. The result: 85.8% fewer parameters than ResNet50, same accuracy.

6.2 MobileNetV1 - Mitigating Vanishing Gradient — Proof

Note: MobileNetV1 does NOT use residual connections (that is MobileNetV2). Vanishing gradient is mitigated by two mechanisms:

Mechanism	How it works	Why it prevents vanishing gradient
BatchNorm after every DSC layer	Normalises each layer output to zero mean, unit variance before passing to next layer	It keeps the activation values in a proper range. When they don't shrink toward zero, the gradients during backprop also stay stable across all 28 DSC layers.
ELU in classification head	Derivative = 1 for x>0, α·e^x > 0 for x<0	The gradient in the head never becomes zero, so neurons don't "die." This keeps a clear path for gradients to flow from the loss back through the head to the backbone features..

Parameter count	Total	Phase 1 Trainable	Phase 2 Trainable
Part B — MobileNetV1	3,509,187	279,683	1,873,539
Assignment 2 — ResNet50	24,705,411	1,116,419	16,092,419
Reduction	85.8% fewer	75.0% fewer	—

7. Part B — Results & Plot Analysis

7.1 Training Curves

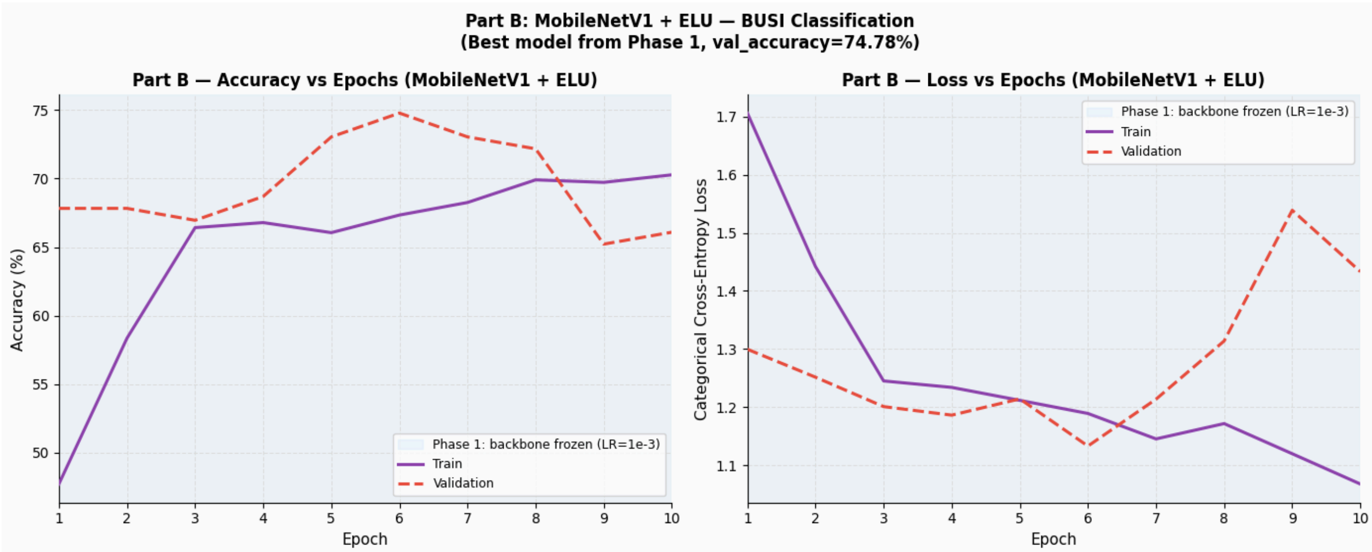

Figure 5: Part B Phase 1 training curves (Phase 1 best model used — val_accuracy=74.78%). Blue shading = Phase 1 (backbone frozen, LR=1e-3). Only Phase 1 plotted because Phase 2 did not improve.

Why validation is above training: Training images receive augmentation every epoch — flips, rotation, noise — making them harder to classify correctly. Validation images are clean (no augmentation). So val accuracy (68% epoch 1) starts above train accuracy (48% epoch 1).

Why validation rises non-uniformly (67.8% → 74.78%): Epochs 1–3 are flat while the head adjusts from random initialisation. Epochs 4–6 show rapid improvement as ELU head converges and class weights balance all three classes. Loss decreases smoothly from 1.70 → 1.13, confirming Cross-Entropy Properties 1 and 3.

Why Phase 2 made things worse: With only 545 training images, unfreezing the last DSC block caused the pretrained ImageNet features to shift without enough data to find better ultrasound features. Val accuracy dropped from 74.78% to 70.43%. Phase 1 best model was used.

7.2 Test Evaluation

Class	Precision	Recall	F1	Support
Benign	86.21%	74.63%	80.00%	67
Malignant	63.89%	71.88%	67.65%	32
Normal	65.38%	80.95%	72.34%	21
Macro Average	71.83%	75.82%	73.33%	120
Overall Accuracy	75.00%	—	—	120

7.3 Confusion Matrix

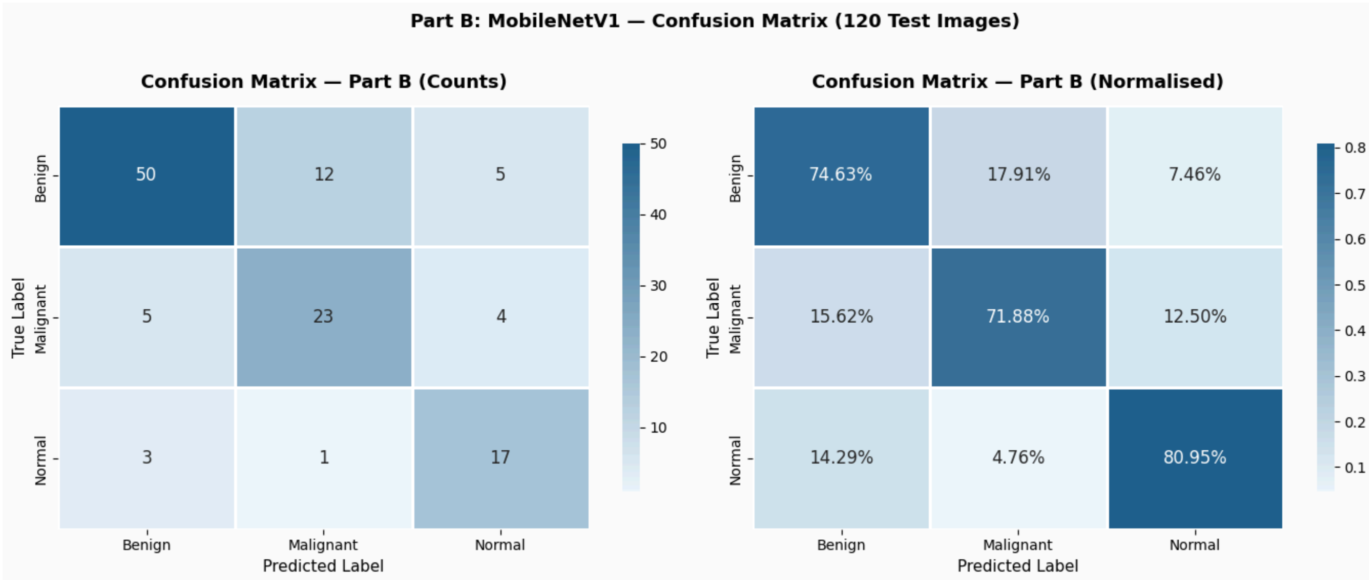

Figure 6: Part B confusion matrix — counts (left) and normalised by true class (right).

Normal (17/21 = 80.95%): Dramatically better than Part A (14.3%) because the full pretrained DSC backbone provides rich features for all 28 layers — far more capacity than Part A's 2 DSC blocks.

Malignant-Benign confusion (5 cases): A genuine clinical challenge — complex benign masses visually resemble malignant lesions.

7.4 Per-Class Metrics

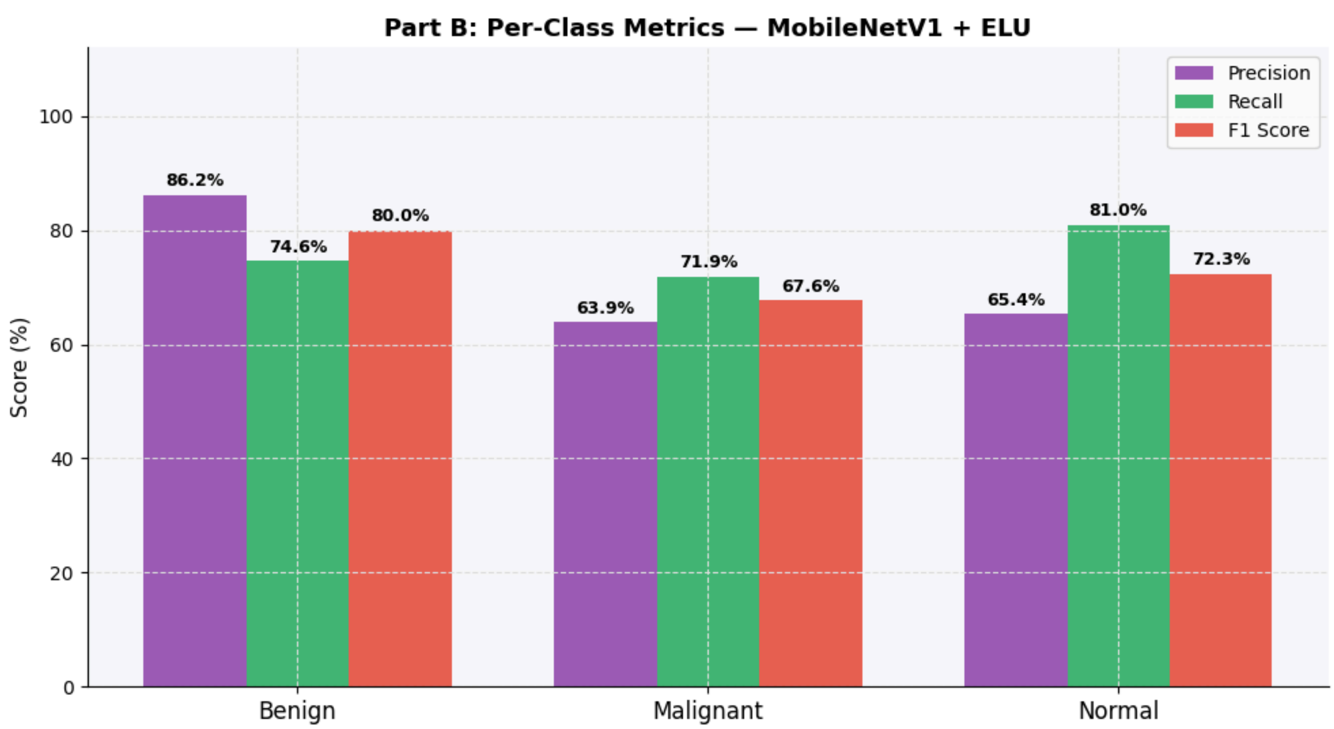

Figure 7: Part B per-class metrics. All three classes have strong performance — Normal F1=72.3%, Malignant F1=67.6%, Benign F1=80.0%.

Why Normal improved from F1=24% (Part A) to F1=72.3% (Part B): Part A had 2 DSC blocks trained from scratch on 92 Normal images. Part B has a full pretrained DSC backbone with rich features that are adapted via the classification head. The full backbone provides far more representational capacity for the Normal class.

8. Comparison — Assignment 2 vs Part A vs Part B

8.1 Summary Table

Model	Accuracy	Macro F1	Trainable Params	Normal Recall	Key difference
Assignment 2 — ResNet50 + ReLU + Dense	75.83%	74.04%	1,116,419	76.2%	Dense head, ReLU — possible dead neurons
Part A — ResNet50 + DSC + ELU	67.50%	55.15%	1,143,939	14.3%	DSC head only — limited capacity for minority class
Part B — MobileNetV1 + ELU	75.00%	73.33%	279,683	80.95%	Full DSC backbone — 85.8% fewer params, same accuracy

8.2 Connecting Results to Loss Function Properties

Property	Evidence from results
1. Monotonically increasing	single minimum — no oscillation.
2. Differentiable (no halting)	Training never halted mid-epoch. EarlyStopping triggered by val_accuracy plateau — not by training failure. Gradients were always defined and computable.
3. Faster convergence	Part A: val loss drops 0.18 in first 2 epochs (fast, large error), then 0.07 in next 4 epochs (slow, small error). Part B: same pattern. Cross-Entropy gradient proportional to error gives fast early updates then gentle fine-tuning.

9. Conclusion

- Categorical Cross-Entropy satisfies all 3 desirable properties — monotonically increasing (Property 1), differentiable with gradient = y_pred − y_true (Property 2), and gradient proportional to error for faster convergence (Property 3).
- ELU is the ideal activation. Derivative always > 0 (no dead neurons, no vanishing gradient). Output centred near zero (faster convergence than Leaky ReLU).
- Class weights (0.5956 / 1.2358 / 1.9534) prevent the Benign majority from dominating the loss, ensuring all three classes converge at similar rates — directly addressing the slower convergence problem for minority classes.

- Part A (DSC head + ELU): 67.50% accuracy. The training loss decreases as expected. Normal recall is only 14.3% — the 2-block DSC head lacks capacity for 92 Normal training images. This is the trade-off: less computation, less capacity.
- Part B (MobileNetV1 + ELU): 75.00% accuracy with 85.8% fewer parameters than ResNet50. Full DSC backbone provides sufficient capacity — Normal recall rises to 80.95%. Same accuracy as Assignment 2 with 85.8% fewer parameters proves DSC is computationally efficient without sacrificing performance when used as a complete architecture.
- Patient hold-out split (545/115/120) ensures zero patient overlap between all splits. Same 120 test patients used for all 3 models

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